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Aim: Develop a web application to fetch real-time weather data for a city. Use AJAX & a public weather API to retrieve information. Display temperature, humidity, and weather conditions dynamically.

Code:

```
<!DOCTYPE html>
<html>
<head>
   <title>Real-Time Weather App</title>
   <meta name="viewport" content="width=device-width, initial-scale=1.0">

<style>
* {
   margin: 0;
   padding: 0;
   box-sizing: border-box;
   font-family: Arial, sans-serif;
}

body {
```

```
    height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;

    /* Ocean blue gradient */
    background: linear-gradient(135deg, #0077b6, #00b4d8, #90e0ef);
}

/* Main card */
.app {
    background: rgba(255, 255, 255, 0.85);
    padding: 30px;
    border-radius: 20px;
    width: 320px;
    text-align: center;
    color: #023047;
    box-shadow: 0 15px 35px rgba(0,0,0,0.2);
}

/* Heading */
h1 {
    margin-bottom: 20px;
    color: #0077b6;
}
```

```
/* Input */
input {
    width: 100%;
    padding: 12px;
    border: 2px solid #90e0ef;
    border-radius: 10px;
    margin-bottom: 12px;
    font-size: 16px;
    background: white;
    outline: none;
}

/* Button */
button {
    width: 100%;
    padding: 12px;
    border: none;
    border-radius: 10px;
    background: #0077b6;
    color: white;
    font-size: 16px;
    cursor: pointer;
    transition: 0.3s;
    font-weight: bold;
```

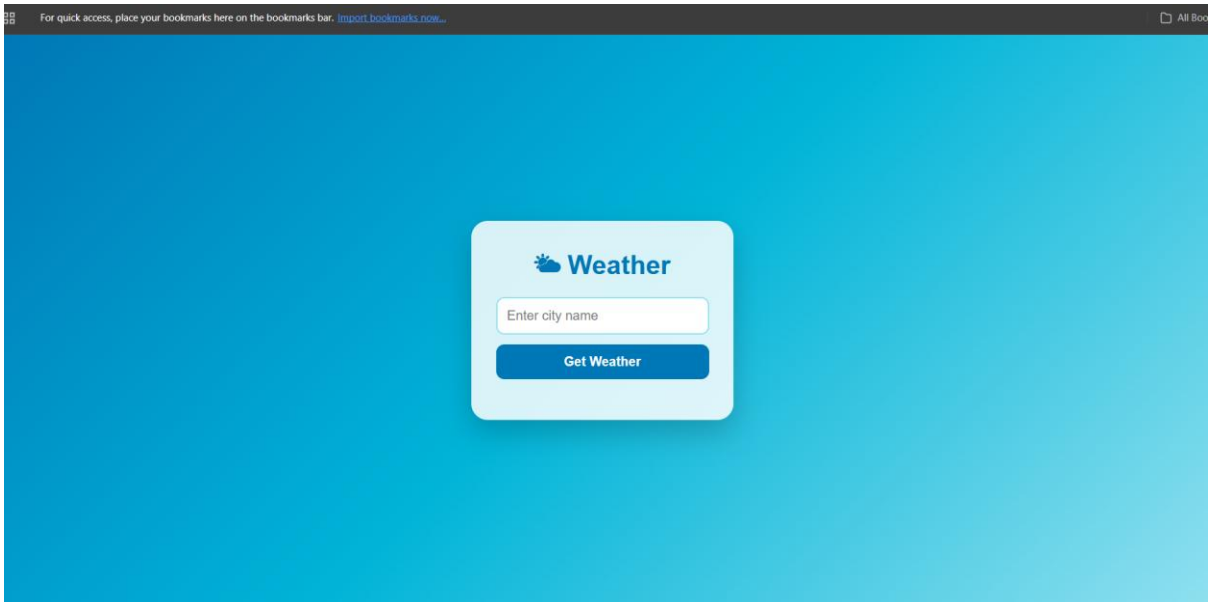
```
}

button:hover {
    background: #023e8a;
    transform: translateY(-2px);
}

/* Weather display */
#weatherBox {
    margin-top: 20px;
}

.temp {
    font-size: 42px;
    font-weight: bold;
    margin: 5px 0;
}

.icon {
    width: 80px;
}
</style>
</head>

<body>
```

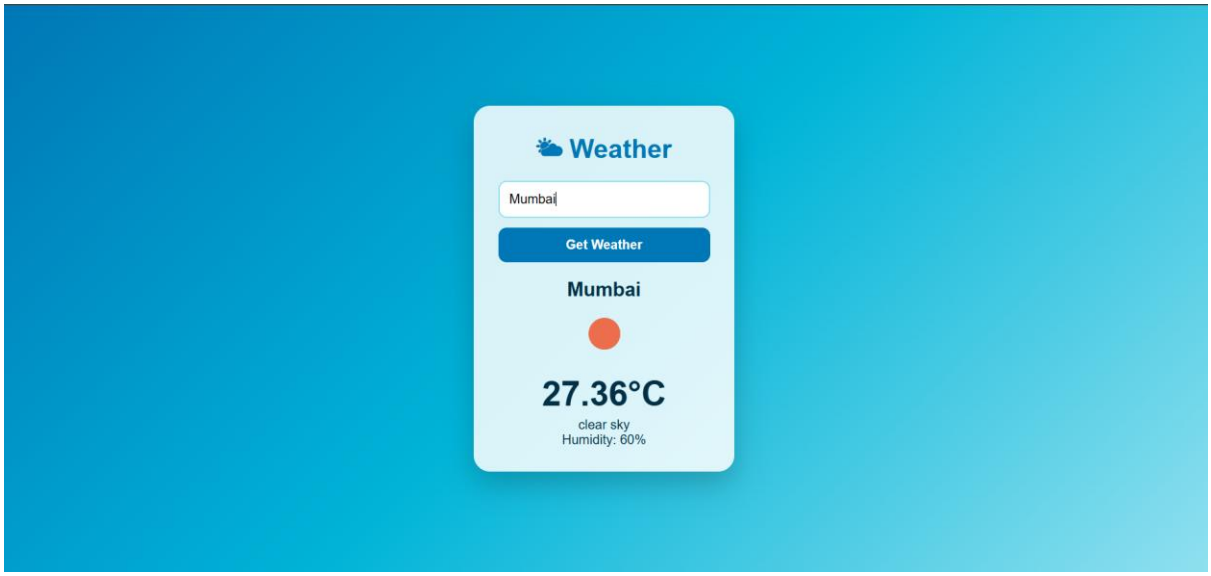
```
<div class="app">
    <h1>⛅ Weather</h1>

    <input type="text" id="cityInput" placeholder="Enter city name">
    <button onclick="getWeather()">Get Weather</button>

    <div id="weatherBox"></div>
</div>

<script>
const apiKey = "0455fee8de3eaddad5d24dcd8730b0c4";

document.getElementById("cityInput")
    .addEventListener("keypress", function(e) {
        if (e.key === "Enter") getWeather();
    });

function getWeather() {
    const city = document.getElementById("cityInput").value;

    if(city === "") {
        alert("Enter a city name");
        return;
    }
```

```
    document.getElementById("weatherBox").innerHTML = "Loading...";

    const url =
`https://api.openweathermap.org/data/2.5/weather?q=${city}&appid=${apiKe
y}&units=metric`;

    fetch(url)
        .then(response => response.json())
        .then(data => {

            if(data.cod === 200) {

                const temp = data.main.temp;
                const humidity = data.main.humidity;
                const condition = data.weather[0].description;
                const icon = data.weather[0].icon;

                document.getElementById("weatherBox").innerHTML = `
                    <h2>${data.name}</h2>
                    <img class="icon"
                        src="https://openweathermap.org/img/wn/${icon}@2x.png">
                    <div class="temp">${temp}°C</div>
                    <p>${condition}</p>
                    <p>Humidity: ${humidity}%</p>
                `;
```

```
        } else {
            document.getElementById("weatherBox").innerHTML =
                "City not found 😢 ";
        }
    })
    .catch(() => {
        document.getElementById("weatherBox").innerHTML =
            "Error fetching data";
    });
}
</script>
</body>
</html>
```

Output:




Weather
Mumbai
Get Weather
Mumbai
27.36°C
clear sky
Humidity: 60%